SWE5001 PRACTICE PROJECT · TEAM 8

BLOCBUY

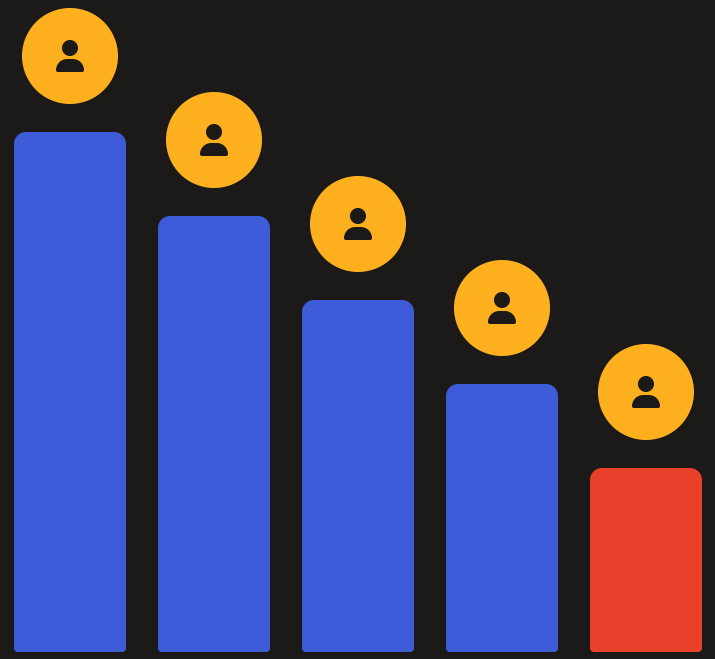
ONE FOR ALL • ALL FOR ONE

A scalable community group-buying platform

Three services • event-driven • rebuilt on every release

Lim Jiechao · Liu Jin · Panpakorn Siripanich · Wang Ziang · Yu Qianshuo

Graduate Certificate in Architecting Scalable Systems · 21 October 2026



The price drops as each member joins

Group buying only works if enough people join in time

Today: chat groups and spreadsheets

- Buyers can't see deadlines or progress
- Organisers collect money and refund by hand
- Sellers have no place for team prices

With BlocBuy

- Pay to join a team
- More members, lower price
- Team fills, or everyone is refunded automatically

JC Lim

A two-sided platform built around one interaction: forming a team

Seed a few curated launch offers

Network effect each new member lowers the price

Producers

- Sellers and admins publish offers
- Buyers who start or share a team

CORE INTERACTION

Form a team at a team price

Consumers

- Browse offers
- Join a team
- Track orders

Consumers become producers when they start or share a team

Shared, reusable services

Identity | Notifications | Payments

Future seeds: event tickets · bulk subscriptions · neighbourhood produce boxes

Scope: the full buying journey, on three services

Identity service

- Sign in with Google
- Buyer and admin roles

Commerce service

- Browse team offers
- Start or join a team, and pay
- Team succeeds, or expires with refunds
- Share invites, order history
- Admin listings

Notification service

- Team succeeded or expired
- Payment received

Not in scope: multi-region · real payments (Stripe test mode) · email or push · extracting payment or catalog into their own services (documented evolution path)

Status: what is verified on AWS, and what is still open

Verified on AWS · 17 Oct

- Built from nothing: 3 services healthy
  commerce 2/2 · identity 1/1 · notification 1/1
- 9/9 smoke checks through CloudFront
- SNS → SQS routing confirmed
  with synthetic events
- Full teardown: 115 resources in 709 s
  absence check clean
- One task load-tested: 200 req/s
  0 errors, p95 < 60 ms (16 Oct)

Verified on 19 Oct

`[FILL 19 Oct: green tick only with evidence in docs/swe5001/evidence/]`

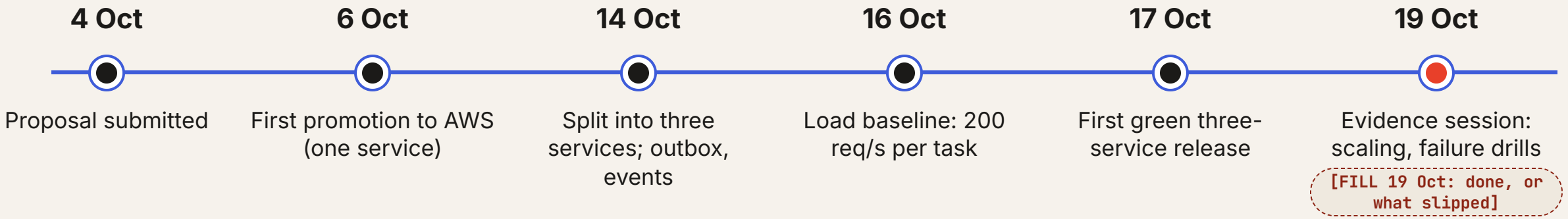
- Auto-scaling under load (V3)
- Queue scaling (V4)
- Self-healing (V5)
- Poison message to DLQ (V7)
- Identity outage drill
- Buyer journey end to end on ECS (#460)

Still open

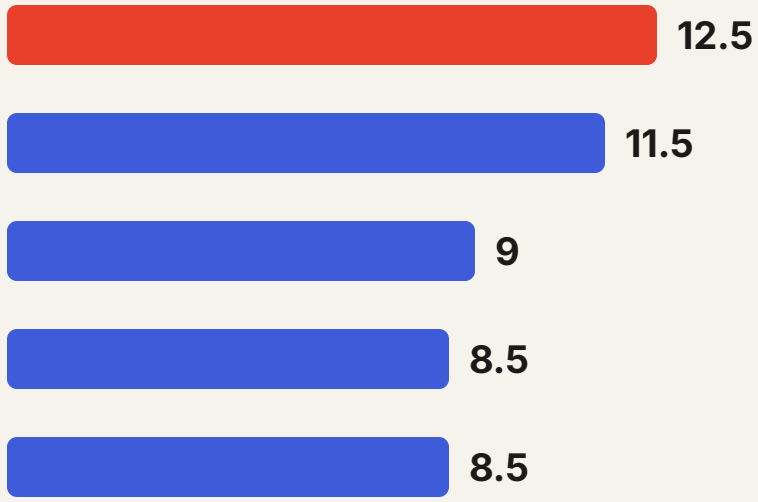
- An interrupted start-up can leave auto-scaling paused
  manual repair in place
  `[VERIFY: fix PR open? if not, no "fix in progress"]`
- Database and cache in one zone
  a cost choice; multi-zone for production
- Alarm email confirmation (#515)

Anything not run on 19 Oct moves here with one honest line, e.g. "V4 queue scaling: verified locally (141 s), not yet on AWS".

On schedule, about 50 person-days across five people



Effort by member (person-days) [CONFIRM: by all five, 18 Oct]



Total ≈ 50 person-days, estimated from the issues each person delivered

WZ Eric

Why three services, not five

AD-02 · ADR 0008

One payment = one database transaction

Payment → Order → Stock → Team

Splitting this needs sagas, so it stays together in **Commerce service**

Identity: used through one small interface

Identity service

Notifications: nothing calls them

Notification service

Our rule

Split only where no transaction crosses

✕ **Rejected: five services**
Sagas everywhere, about 50–55 person-days: our whole budget

WZ Eric

Four decisions shape the logical design

AD-02

Split by transaction boundary

No distributed transactions to build or debug

AD-04

Events through an outbox and queues

An event is sent only if the change really committed

AD-05

One shared login session

Any service checks roles without calling identity

AD-06

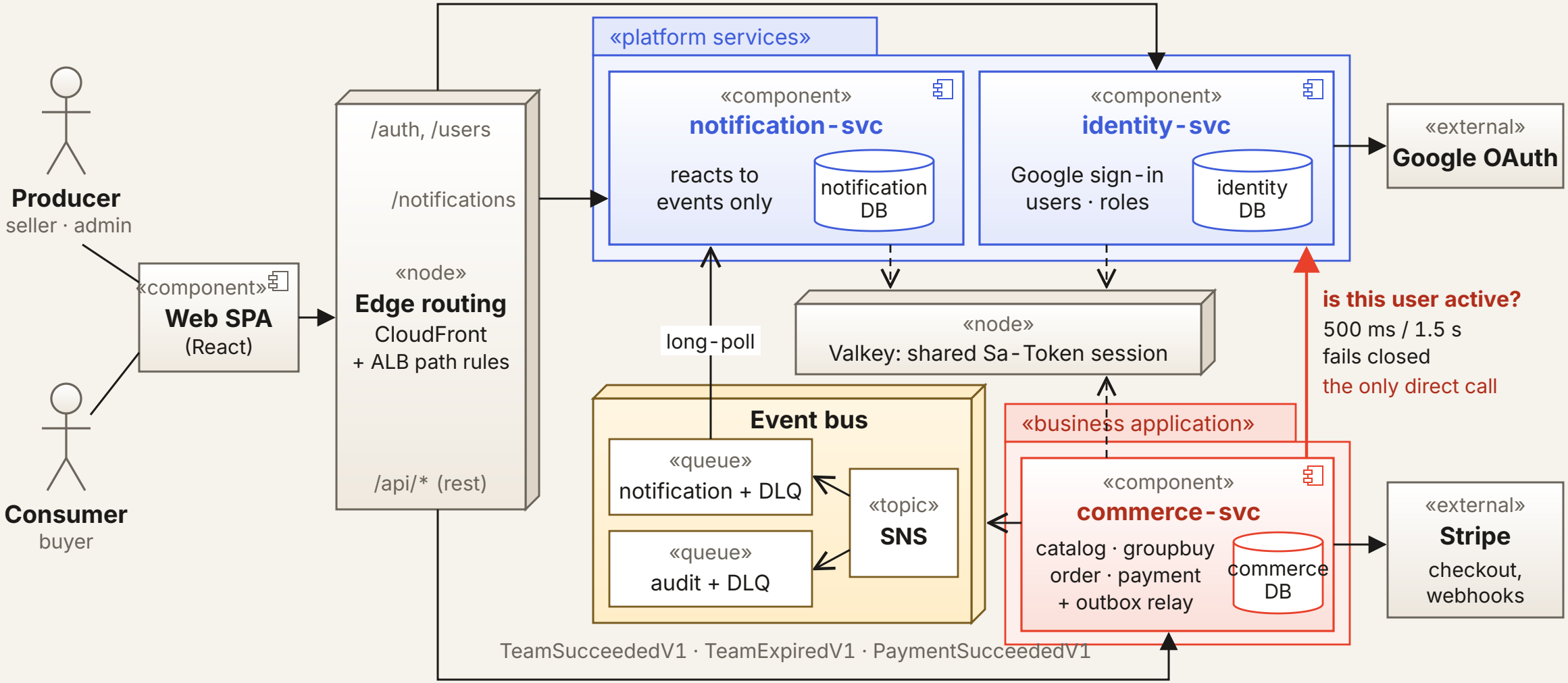
Fail safe when identity is down

Purchases are refused, never half-done

Cost of these choices: notifications lag about 2 s (relay interval); purchases pause during an identity outage

WZ Eric

Logical view: one direct call, everything else by events



One direct call (commerce → identity); everything else by events.

WZ Eric

Bounded contexts decide the services; events are the contract

Commerce service

Catalog
products, team prices

Group buying
teams, members

Orders
orders

Payments
payments, refunds

Identity service

Identity
users, roles

Notification service

Notification
notifications

Domain event → integration event

TeamSucceeded	TeamSucceededV1
TeamExpired	TeamExpiredV1
PaymentSucceeded	PaymentSucceededV1

- Inside a service: free to change
- Between services: versioned, IDs only

[CONFIRM: Option A (this) or B: context map, Eric]

WZ Eric

How producers and consumers use the platform, and what reuse costs

Producers

- Sellers and admins publish listings
- Buyers start and share teams

Identity
one account for both sides

Commerce
offers, teams, payments

Events
any new consumer just subscribes

Notification
tells people what happened

Consumers

- Browse and join
- Track orders
- Get notified

Measured reuse

`[FILL 19 Oct: from PR; drop row 1 if not found by Tue]`

Added to the platform	What it took	Commerce changes
Audit consumer	`[FILL 19 Oct: N]` lines in `[FILL 19 Oct: M]` files, PR #`[FILL 19 Oct: …]`	0
A new service	One entry in the Terraform services map (`ecs-service` golden-path module)	0

One public contract

`contracts/openapi.json` → generated TypeScript client
Events versioned (V1), IDs only

Every new service starts with

Login & roles · Error format · Tracing · Outbox · Duplicate protection

`blocbuy-platform-starter`

LJ Liu Jin

Physical decisions: managed, disposable, low cost

Fargate containers on ARM

No servers or cluster to run · ~20% cheaper · EKS control plane alone ≈ US$73/month

AD-07 · 13

Disposable environment

Built per session or release, then destroyed

AD-12

One database server

A database per service: isolation at low cost; own instance is the scale-out path

AD-11

Small NAT instance

≈ US$4 instead of US$43 a month

AD-10

No stored cloud keys

Short-lived pipeline tokens · secrets in Parameter Store

AD-14

MEASURED, NOT ESTIMATED

US$0.49

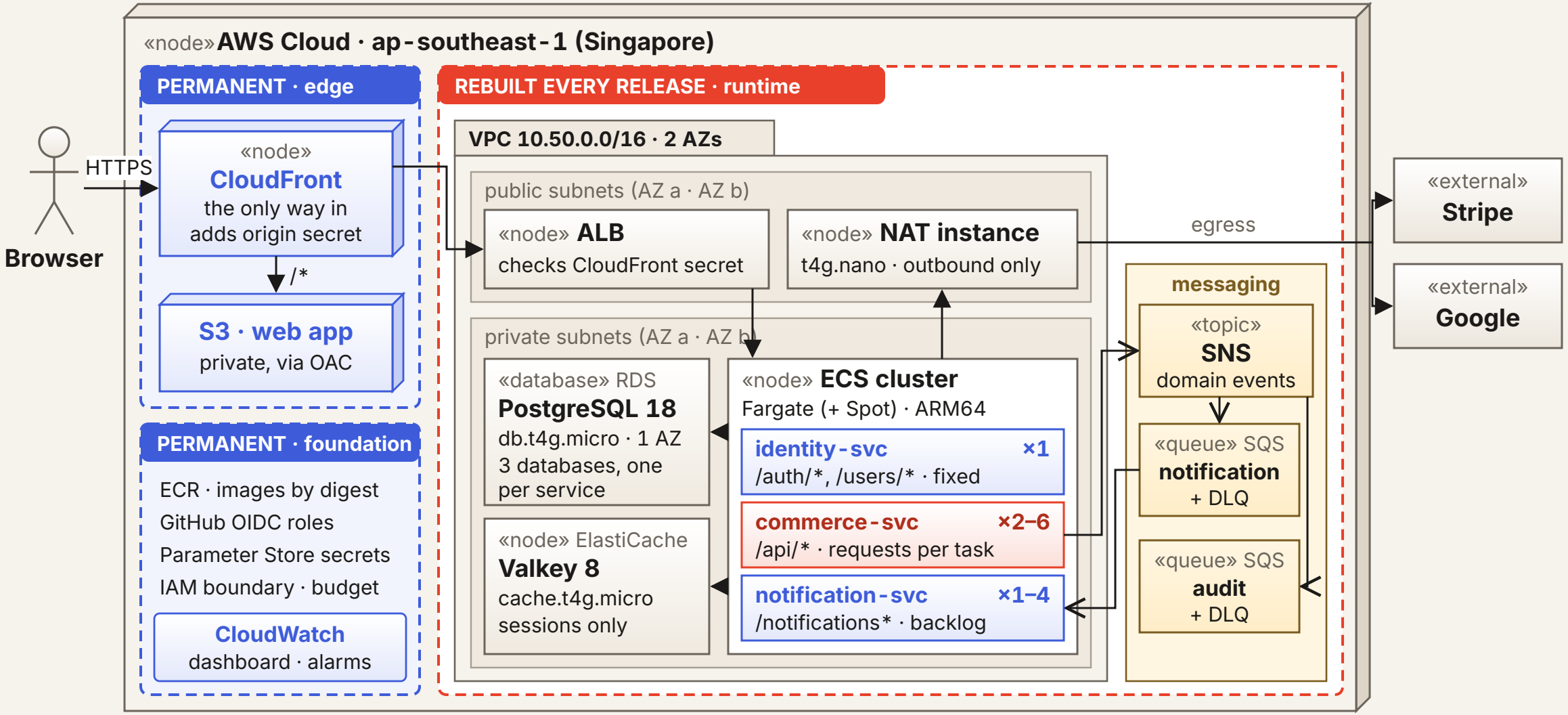
in total for the runtime's database, cache and containers across **7 up/down sessions**, 15–17 Oct

AWS Cost Explorer, filtered to the runtime services

Trade-off: less redundancy (one NAT, single-zone database), acceptable for this project

LJ Liu Jin

Physical view: private services behind CloudFront, across two zones



LJ Liu Jin

Each service scales on its own signal, and how far it goes

Commerce
ALB requests per task (target 12,000/min) → **2 to 6 tasks**

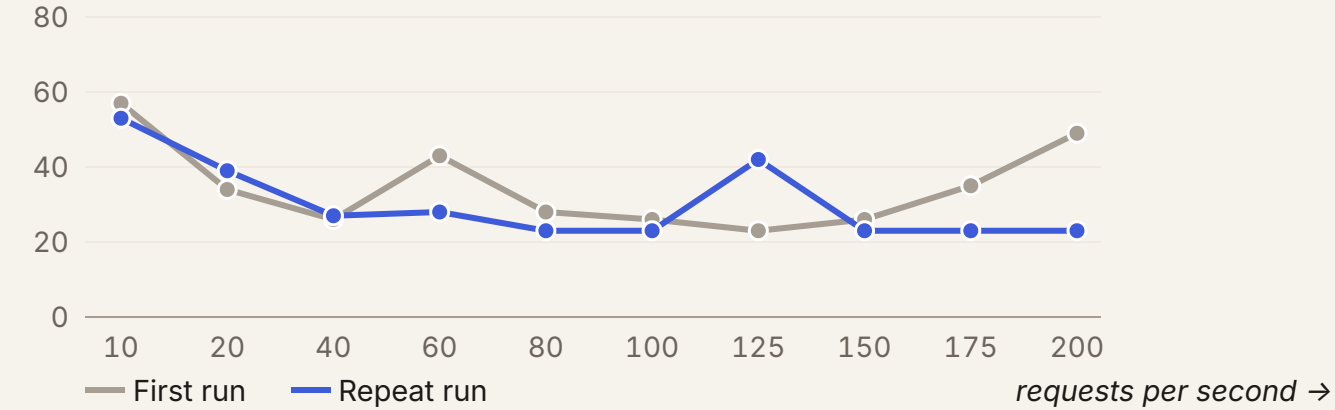
Notification
SQS backlog per task (target 50) → **1 to 4 tasks**

Identity
Fixed at **1 task**

Capacity statement

Measured: `[FILL 19 Oct: req/s, three services]`
Projected ceiling at max tasks: ≈ 6 × 200 = **1,200 req/s** for browse **(projection, not measured)**
Next bottleneck: the single database and the single outbox relay
`[VERIFY: ~50 events/s measured or estimated?]`

Load test, one task, 16 Oct: response time (p95, ms)



200 req/s	**< 60** ms	**0** errors
highest tested, not saturated	95% of responses	57,600 requests

Three-service shape, 19 Oct (V3):
`[FILL 19 Oct: N tasks served X req/s at p95 Y ms, Z% errors]`
Place the CloudWatch crop here: “Services: task count” + “Edge: ALB latency” over the V3 window. If V3 did not run, delete this box.

LJ Liu Jin

Each service owns its data



Each service reads only its own tables

per-service roles; enforced by grants and tested

The cache holds sessions only

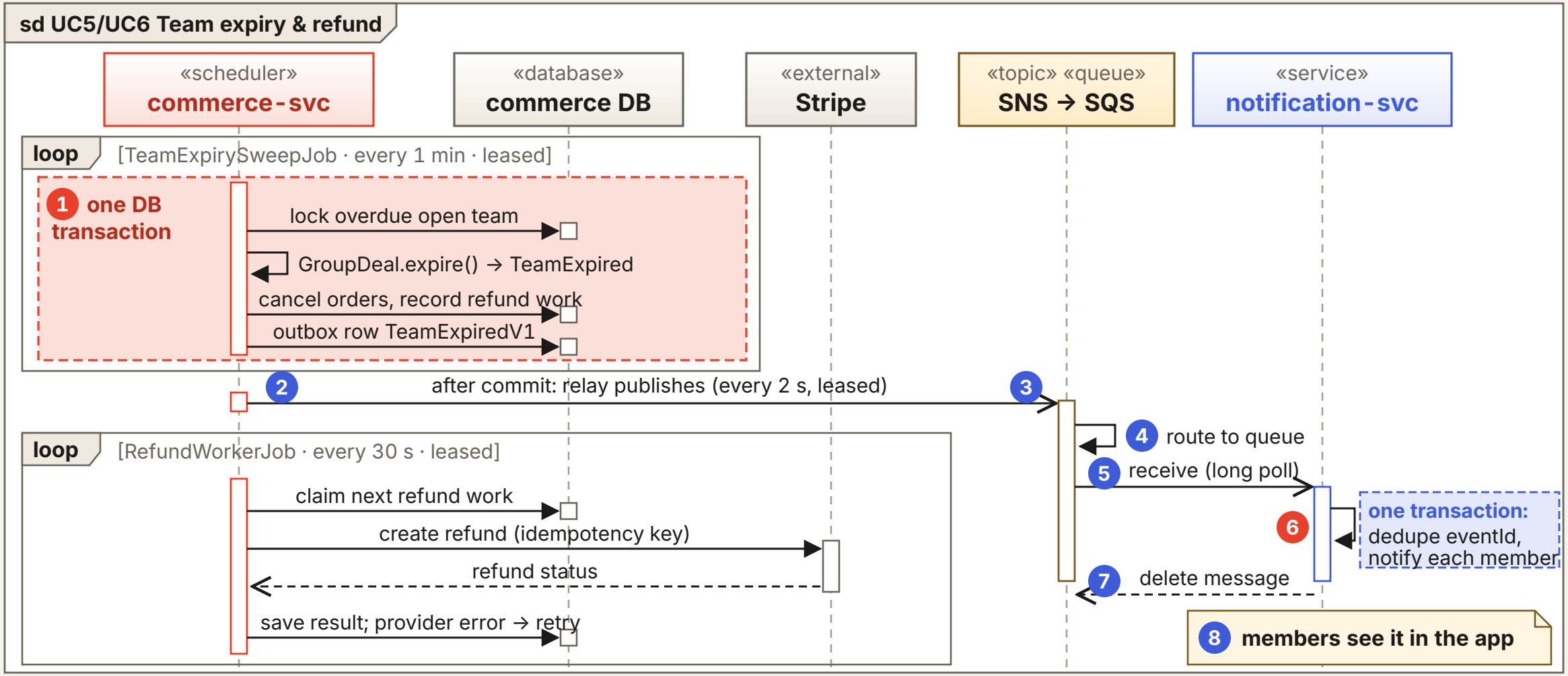
No card data stored; events carry IDs only

Concurrency

Row locks in a fixed order (**team → intent → payment**) settle the last-seat race. Scheduled jobs hold a database lease, so one task runs them however many are up.

YQ Yorik

Team expiry, end to end: one transaction in, one transaction out



[VERIFY: step 8: each member or team owner?]

[CONFIRM: swap go/no-go after Yorik rehearses; fallback B15]

YQ Yorik

Failures are expected, and handled



If...

A message arrives twice

Ignored: its (event ID, consumer) is already recorded



If...

A message keeps failing

Moved to a dead-letter queue after 5 tries; an alarm fires



If...

Publishing fails

The event stays in the outbox and is retried



No lost events

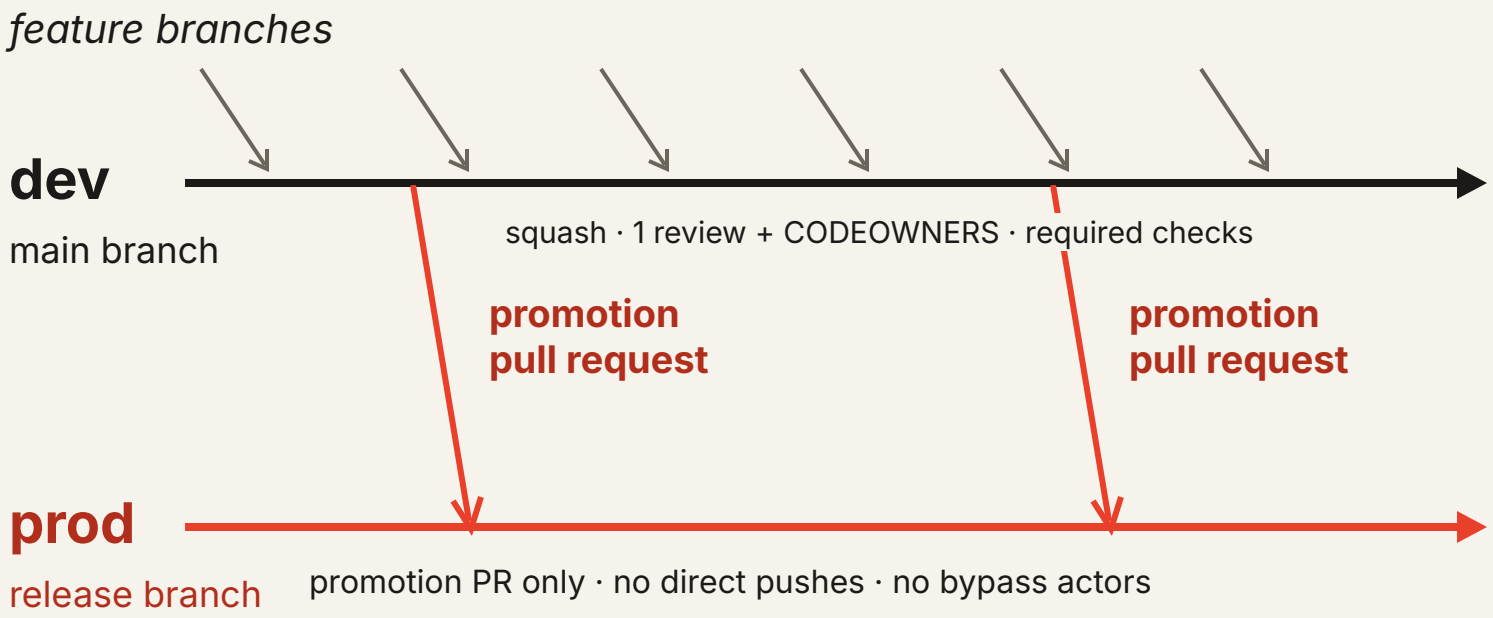


No double notifications



Bad messages isolated

One main branch; production changes only by promotion



Review + automated checks

No direct pushes to prod

Artefacts

Container images
one ARM64 image per service, stored by digest, not tag

Release record
which digests are live

API contract
one file in the repository

Every pull request is checked before it can merge

Build & test

- Unit tests and API contract check
- ≥ 80% coverage on new code (SonarCloud)

Container images

- Build every service
- Boot each one [VERIFY]

Security

- Secret scanning (gitleaks)
- Dependency and image scanning (Trivy)
- Static analysis [VERIFY: CodeQL runs?]

Infrastructure

- Terraform fmt, validate, test
- Policy checks (checkov)
- Plan posted to the PR

5,252 workflow runs, 16 Sep – 16 Oct

94% / 95% backend / frontend green

8.6 / 3.1 min median run, backend / frontend

If any check fails, the pull request cannot merge

One promotion builds, deploys, tests and releases, then destroys it all

Approve
promotion pull request

Build
container images
[VERIFY: run log]
first run 6 Oct: 2.5 min

Create + deploy
environment and smoke tests
≈ 32 min
first run 6 Oct: 13 min

Release
mark images as live
[VERIFY: run log]
first run 6 Oct: 1 min

Destroy
and check nothing is left
≈ 12 min
first run 6 Oct: 10 min

51 min end to end, three services, 17 Oct (#534)

28 min on 6 Oct with one service. Includes queueing and approval gaps between jobs [VERIFY]. 115 resources destroyed in 709 s.

Destroy always runs
even if an earlier step failed

Failed deploy rolls back
the ECS circuit breaker restores the last good version

Release digests in Parameter Store are the last-known-good set

Every release is a rebuild from zero, so every release also tests disaster recovery

What the real cloud taught us

1

Mocks passed, the real deploy failed

Missing cloud permissions only showed up in a real run (#527)

Test permissions for real

2

Events lost right after deploy

New SNS subscriptions take up to 15 minutes to route; nothing errors, messages just go nowhere

Deploy now waits for routing

3

Auto-scaling silently paused

A retried start-up reported success but left scaling suspended

Check scaling, don't assume it

No standing keys: who and what can touch AWS

People
IAM Identity Center

admins (3) · developers (2)
1-hour sessions

Machines
GitHub OIDC

separate plan / deploy / apply roles
no AWS keys in GitHub

Disclosed exception
2 IAM users for SSM port forwarding (1 active key), bounded by the SCPs

AWS Organization management account runs no workloads

Root SCP · Org-Baseline SCP · Region-Pin: ap-southeast-1 + global

Sandbox OU SCP · Sandbox-Guardrails
no expensive services · small instance sizes only

team8 project account

SWE5006 · dev
EC2 host

SWE5001 · prod
ECS runtime, rebuilt per release

Guardrails above the account: SCPs bind even the account's admins and root user

Live demo

1 **Buy as a customer, across three services**

Functionality

2 **Auto-scaling under a load test**

Scalability, performance

3 **Stop a server, nothing breaks**

Availability

4 **Identity goes down, payments stay safe**

Consistency, security

5 **Event backlog and the dead-letter queue**

Scalability, reliability

6 **The pipeline that deployed it**

Maintainability

- **Load test running since 16:28 · everything you see is live on AWS** `[CONFIRM: else "Live on AWS; recorded steps are marked"]`

Thank you

Questions?

WHAT WE HOPE YOU REMEMBER

- Split where no transaction crosses
- Events never lost or doubled
- Fails safe when identity is down
- Rebuilt from scratch every release

[FILL 19 Oct: N tasks at X req/s]
p95 [FILL 19 Oct: Y] ms (V3)

US$0.49
runtime database, cache and containers, 7 sessions

51 min
approved PR → released, tested, destroyed

github.com/PPsyrius/iss-groupbuy-team8 · d3lvq02o2e7tcn.cloudfront.net

Backup slides: index

B1a **Architecture decisions AD-01 to AD-06 (logical)**
Why X over Y (logical)

B1b **Architecture decisions AD-07 to AD-14 (physical)**
Why Fargate, NAT instance, one RDS, accounts

B2 **Known limitations AISS-01 to AISS-06**
Single points of failure, coarse commerce, /internal

B3 **Context map**
What DDD actually changes

B4 **Aggregates and state machines**
GroupDeal / Order lifecycle

B5 **Layers and enforced boundaries**
How boundaries are kept from eroding

B6 **UC4: join a team by paying**
The payment path in detail

B7 **Technology and services**
Exact services, sizes, versions

B8 **Full findings list**
What else went wrong

B9 **Infrastructure as code and actual bill**
Stacks, lifetimes, real cost

B10 **API access for producers and consumers**
Platform Engineering: robust API access

B11 **Analytics: what the platform measures**
Analytics success criterion

B12 **Scale-out and resilience results**
Evidence behind slides 5, 14, 17

B13 **Logical view, simplified**
Plainer companion to slide 9

B14 **Physical view, simplified**
Plainer companion to slide 13

B15 **Team expiry, numbered steps**
Plainer companion / fallback for slide 16

Click a row (or type its number in the URL as #B6). Slide 24 stays up during Q&A; open a backup only when a question needs it.

WZ Eric

Architecture decisions AD-01 to AD-06 (logical)

ID	Decision	Trade-off / rationale
AD-01	Strangler, not big bang: one codebase, two topologies (monolith on dev, distributed on ECS)	Commerce code churned in 46 commits in Sep → split by assembly, not relocation (ADR 0008)
AD-02	Split along transactional consistency boundaries, not nouns	Settlement and expiry stay single-transaction → no sagas. 5-service split rejected (50–55 person-days)
AD-03	Database per service, role per service	No cross-context joins; enforced by grants + ArchUnit
AD-04	Transactional outbox → SNS → SQS + DLQ, idempotent consumers (ADR 0007)	At-least-once + dedupe = effectively-once; no broker to operate
AD-05	Shared Sa-Token session in Valkey, roles in session	No per-request call to identity for authorisation
AD-06	One sync call commerce → identity, fail closed	Outage → 500 → Stripe redelivers; never wrongly rejects a buyer

LJ Liu Jin

Architecture decisions AD-07 to AD-14 (physical)

ID	Decision	Rationale
AD-07	ECS Fargate ARM64, not EKS	EKS control plane ≈ US$73/mo, more than all Fargate compute; ARM ~20% cheaper than x86
AD-08	Fargate base + Spot (1 : 3) above the minimum	Cheap burst capacity, baseline protected
AD-09	CloudFront → ALB with origin-secret header; ALB default 403	TLS at the edge; ALB unusable except through CloudFront
AD-10	fck-nat t4g.nano instead of NAT Gateway	≈ US$4 vs ≈ US$43 per month
AD-11	One RDS instance, three databases	Logical isolation at demo cost; own instance is the scale-out path, no code change
AD-12	Persistent foundation/edge, disposable runtime	≈ US$0.12–0.20/h while up vs ≈ US$125/mo 24×7 (proposal estimate)
AD-13	No K8s / Argo CD	8–12 days of setup before any feature work (study 03)
AD-14	Dedicated member account in an AWS Organization; Identity Center for people, OIDC for CI, SCP guardrails	Isolated blast radius and billing; short-lived credentials; limits the account admin can't lift

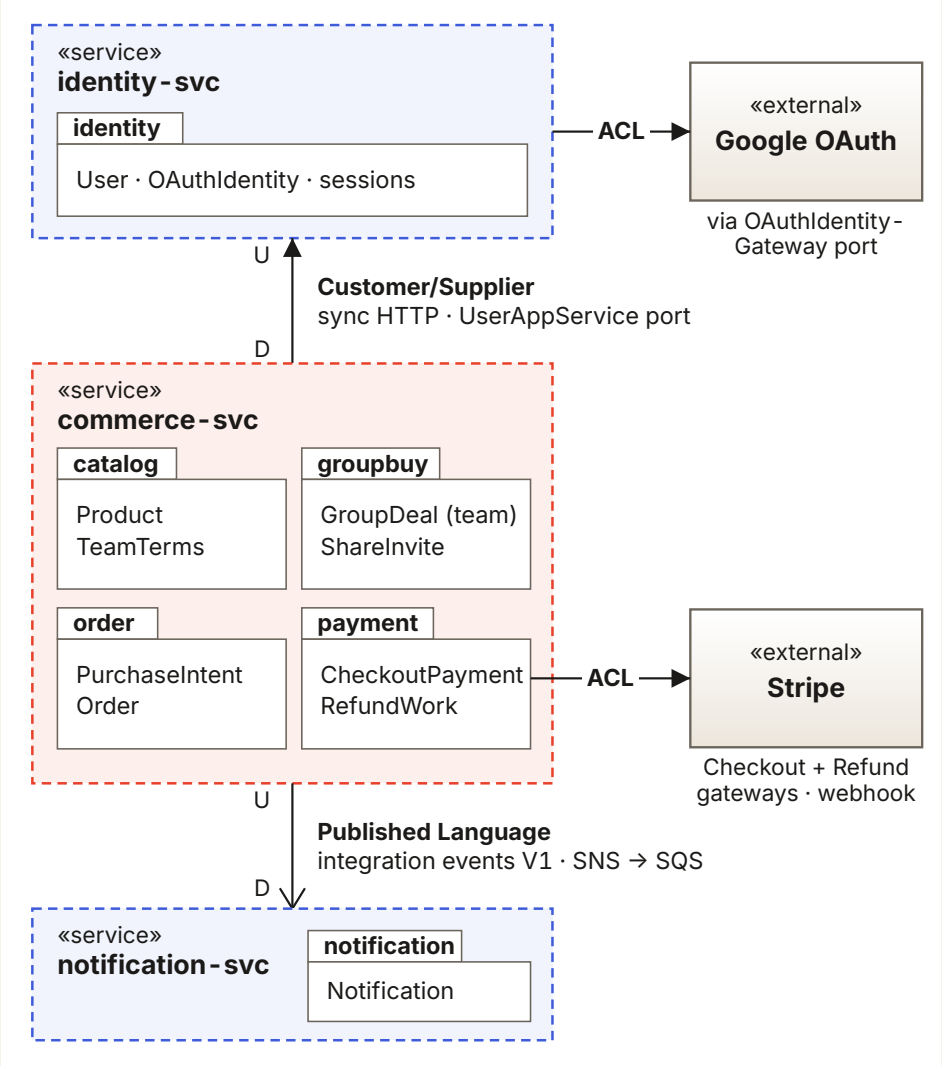
LJ Liu Jin

Known limitations AISS-01 to AISS-06

ID	Limitation	Resolution path
AISS-01	RDS single-AZ, single Valkey node, one fck-nat	Cost-driven; Multi-AZ, replicas, NAT per AZ
AISS-02	commerce-svc is coarse	Extract payment next, with a saga for fulfilment/refund
AISS-03	Identity outage fails checkout closed	By design; Stripe redelivery; cache or circuit breaker later
AISS-04	/internal/** relies on network isolation only	Add a service token or mTLS
AISS-05	One shared RDS instance	Move a DB to its own instance, no code change
AISS-06	Distributed cold start ≈ 32 min (17 Oct, ×2), 15 min of it the SNS filter wait	Pre-warm on the day; edge stays persistent

WZ Eric

Context map



- **identity → Google:** anti-corruption layer via the OAuthIdentityGateway port
- **identity → commerce:** Customer/Supplier, sync HTTP through the UserAppService port (the one direct call)
- **commerce → Stripe:** anti-corruption layer (Checkout and Refund gateways, webhook)
- **commerce → notification:** Published Language, integration events V1 over SNS → SQS

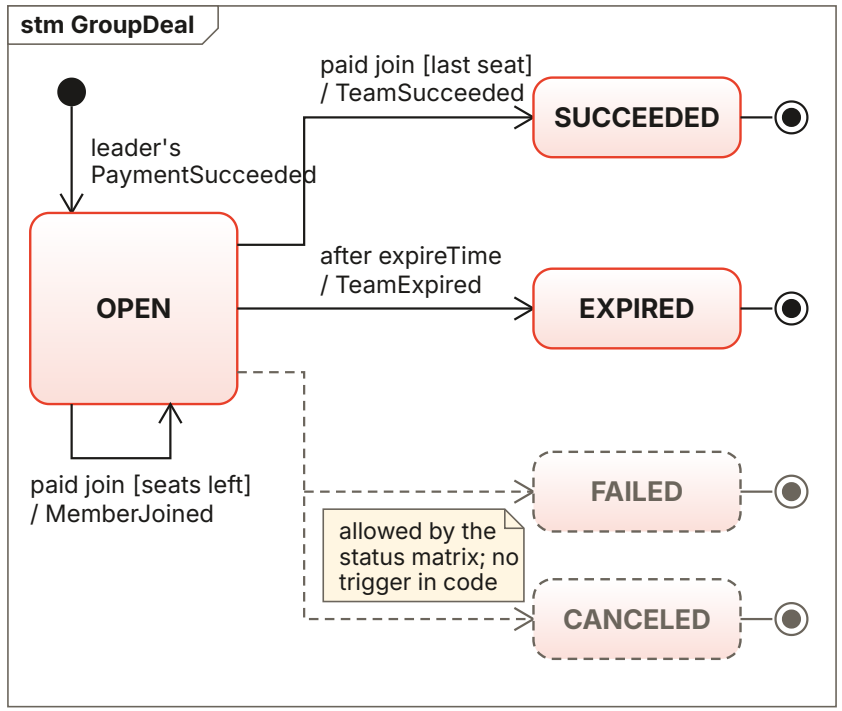
Four contexts share one service because they share transactions; the map shows where the network boundaries are allowed to be.

WZ Eric

Aggregates and state machines

- **GroupDeal (team):** only OPEN may move, once
- **Order:** PENDING → CONFIRMED | CANCELLED
- **PurchaseIntent** is the order-side aggregate (ADR 0001)
- Also: CheckoutPayment, RefundWork, Product + TeamTerms, Notification
- Domain events: MemberJoined, TeamSucceeded, TeamExpired, PaymentSucceeded
- In-process subscribers run before commit (ADR 0002), so in the same transaction the outbox writer maps them to versioned integration events

Be ready to explain: FAILED and CANCELED are allowed by the status matrix but have no trigger in code.



WZ Eric

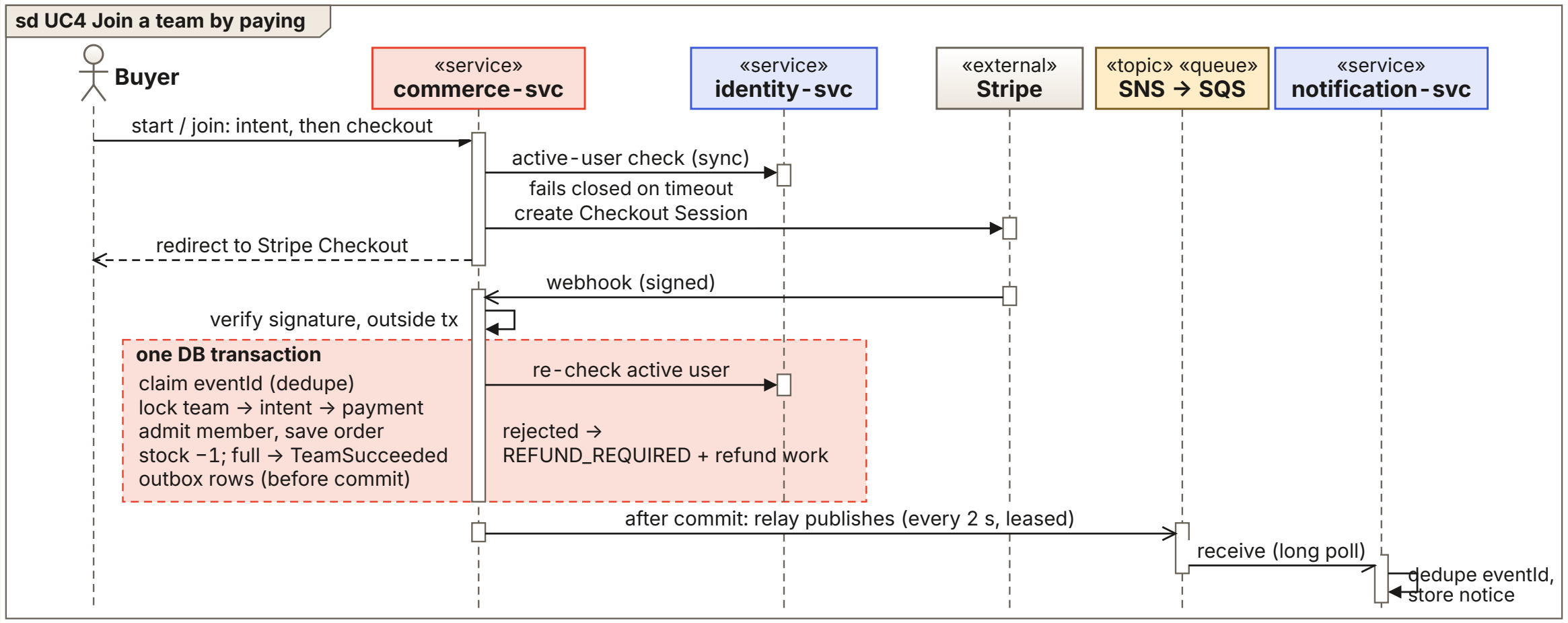
Layers and enforced boundaries

Layer	Module	Contains
Client	`blocbuy-client`	DTOs, integration event contracts (TeamSucceededV1, TeamExpiredV1, PaymentSucceededV1)
Adapter	`blocbuy-adapter`	REST controllers, Result envelope, GlobalExceptionHandler
Application	`blocbuy-application`	Commands, queries, app services, outbox relay
Domain	`blocbuy-domain`	Aggregates, domain events, ports (framework-free)
Infrastructure	`blocbuy-infrastructure`	MyBatis, Stripe, Redis, SNS/SQS transports
Assemblies	`services/*-svc`	Thin Spring Boot entry points that select contexts

Boundaries are tested: 18 ArchUnit rules, e.g. `commerce_reaches_identity_only_through_user_port`, `nothing_depends_on_notification`

YQ Yorik

UC4: join a team by paying



Signature verified outside the transaction; one transaction for dedupe, lock order, admit, order, stock and outbox; the active user is re-checked inside it.

LJ Liu Jin

Technology and services

Concern	Choice
Edge	CloudFront (persistent: OAuth redirect and Stripe webhook depend on its domain), S3 SPA bucket
Compute	ECS Fargate ARM64, 0.5 vCPU / 1 GB, circuit breaker + rollback, Service Connect
Scaling	commerce: ALB requests/target, min 2 / max 6, 60 s out / 180 s in · notification: SQS backlog per task (target 50), 1–4 · identity: fixed 1
Data	RDS PostgreSQL 18 db.t4g.micro (TLS forced) · ElastiCache Valkey 8 cache.t4g.micro (TLS + AUTH)
Messaging	SNS (KMS) → SQS (visibility 60 s) + DLQ (maxReceive 5), filter policy
Secrets	SSM Parameter Store SecureString; DB passwords via write-only arguments (never in state)
Observability	CloudWatch dashboard (Edge · Services · Messaging), DLQ > 0 and queue-age alarms, X-Trace-Id
App stack	Java 21, Spring Boot 4, MyBatis, Sa-Token, Stripe · React 19, TanStack, HeroUI v3

PS Pan

Full findings list

Finding	Fix / status
First promotion: 5 failed runs (digests, tfvars, SG tag deny, early 503)	#405–#417; green 6 Oct
Fargate read-only root FS crashed Tomcat on /tmp	VOLUME ["/tmp"]
Tagging API returns already-deleted ARNs	Per-type describe checks in env-status
Flyway default privileges exposed tables to all roles	afterMigrate grants + ServiceGrantsIT
Outbox relay marked unsent rows failed on a mid-batch error	Caught in review (#485)
Consumer flag never set on ECS → events unconsumed	#512
RegisterScalableTarget AccessDenied: default tags also need TagResource	#527, scoped to Course=swe5001; in effect 17 Oct
Fail-closed messaging gate stopped up twice: alarm email unconfirmed (#520, #528)	#529 testing opt-out; green 17 Oct. Confirm emails, opt-out off (#515) `[FILL 19 Oct]`
Interrupted bootstrap restores suspended scaling	Manual repair only `[FILL 19 Oct: fix status]`
SCP denied CloudFront metrics (us-east-1)	#530: SCPs consolidated, scoped exception

PS Pan

Infrastructure as code and actual bill

Stack	Lifetime	Contents
foundation	persistent	state bucket, GitHub OIDC roles + permissions boundary, ECR, SSM
edge	persistent	CloudFront + S3 SPA (origin switched to the new ALB on each up)
runtime	disposable	VPC, ECS, ALB, RDS, Valkey, SNS/SQS, dashboard, alarms: 113–115 resources

- `env-up.sh / env-down.sh / env-status.sh` (absence check with per-type describe calls)
- Up ≈ 13–15 min (monolith), ≈ 32 min (distributed, 17 Oct ×2); down ≈ 9–13 min (115 resources in 709 s on 17 Oct)
- Golden-path `ecs-service` module: a new service is one map entry
- Apply-role guardrails: IAM only under `/swe5001/`, tag- and name-conditioned denies

Actual AWS bill (whole account, Cost Explorer): Sep US$78.51; 1–17 Oct ≈ US$65 (≈ US$4/day), mostly the always-on SWE5006 EC2 host + ALB. The SWE5001 runtime's RDS + Valkey + ECS: **US$0.49** across 7 up/down sessions (15–17 Oct).

WZ Eric

API access for producers and consumers

Producers & consumers
web SPA, future partner clients

→

CloudFront
the only way in; adds origin secret

→

ALB path rules
/auth, /users, /notifications, /api/*; default 403

→

Services
identity · commerce · notification

Built

- Public contract `contracts/openapi.json`, committed; TypeScript client generated, not committed; contract check on every PR
- Integration events versioned (...V1); consumers ignore unknown versions; producers can add V2 without a lock-step release
- `/internal/**` not routed by the ALB, peer security groups only, absent from the public contract (InternalPathsContractTest)

Not built (be upfront)

- API gateway, per-client keys, rate limiting
- Service-to-service token or mTLS (AISS-04)
- **Path:** API Gateway, or ALB + WAF rate rules behind the CloudFront origin; a service token on `/internal/**`

JC Lim

Analytics: what the platform measures

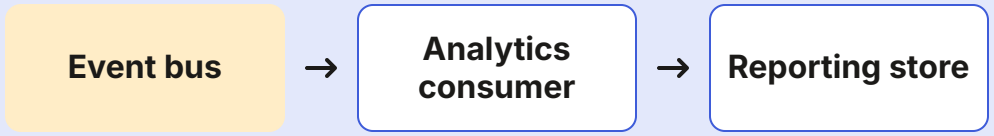
Operational, in use

- CloudWatch dashboard `blocbuy-swe5001`: edge, services, messaging
- V2 load results set the 12,000 req/target/min scaling target
- DLQ and queue-age alarms

Business, in the data, not built

- Team fill rate and time-to-fill per product
- Referral attribution from share invites (UC7)
  `[VERIFY: data in biz_* tables?]`
- Conversion from team start to success; refund rate by product

NEXT STEP



`[FILL 19 Oct: dashboard crop]`
Crop of dashboard **blocbuy-swe5001**, all three rows, from the 19 Oct session.

A read-only consumer on the same bus, the same pattern as audit: no change to commerce.

JC Lim

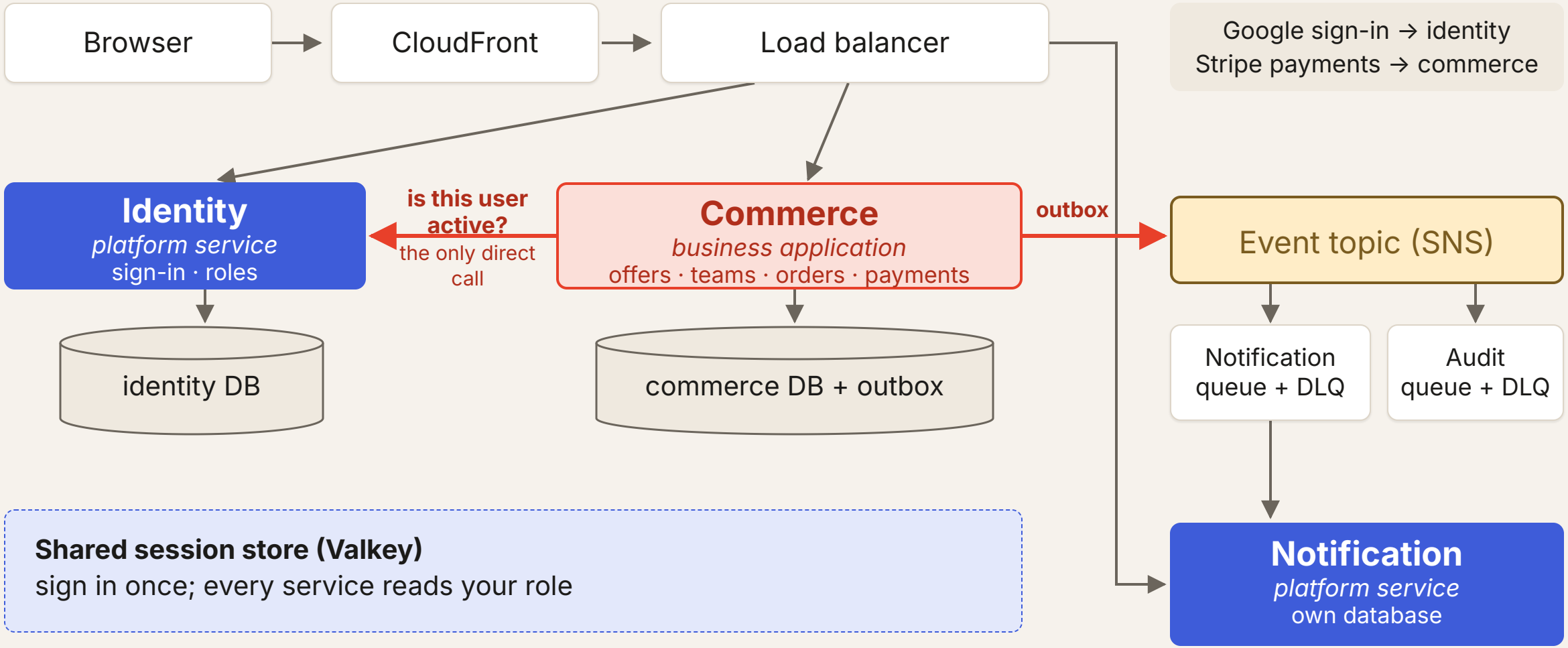
Scale-out and resilience results

Check	Target	Result
V3 HTTP auto-scale	≥ 80% of linear from 1 → 4 tasks; p95 < 800 ms outside 3 min after scale-out	[FILL 19 Oct]
V4 queue scaling	2,000 TeamExpired → 4 tasks, drain, DLQ 0	[FILL 19 Oct] local IT: 141 s end to end
V5 self-heal	stop 1 of 2 tasks, 0 non-2xx, replacement < 2 min	[FILL 19 Oct]
V7 poison message	5 receives → DLQ → alarm	[FILL 19 Oct] local: verified 15 Oct
Identity outage	browse 200; checkout 500 → redelivered once	[FILL 19 Oct] local: verified
Distributed up + smoke	3 services stable, 9 smoke checks	17 Oct ×2: tasks 2/2 · 1/1 · 1/1; 9/9 checks; SNS→SQS verified with synthetic events. Buyer E2E (#460): [FILL 19 Oct]
Teardown	all resources destroyed, absence check clean	115 resources, 709 s (17 Oct); 113, 784 s (16 Oct)

Anything not run on AWS stays "not run on AWS; local result: ...". Never upgrade a local result to an AWS one.

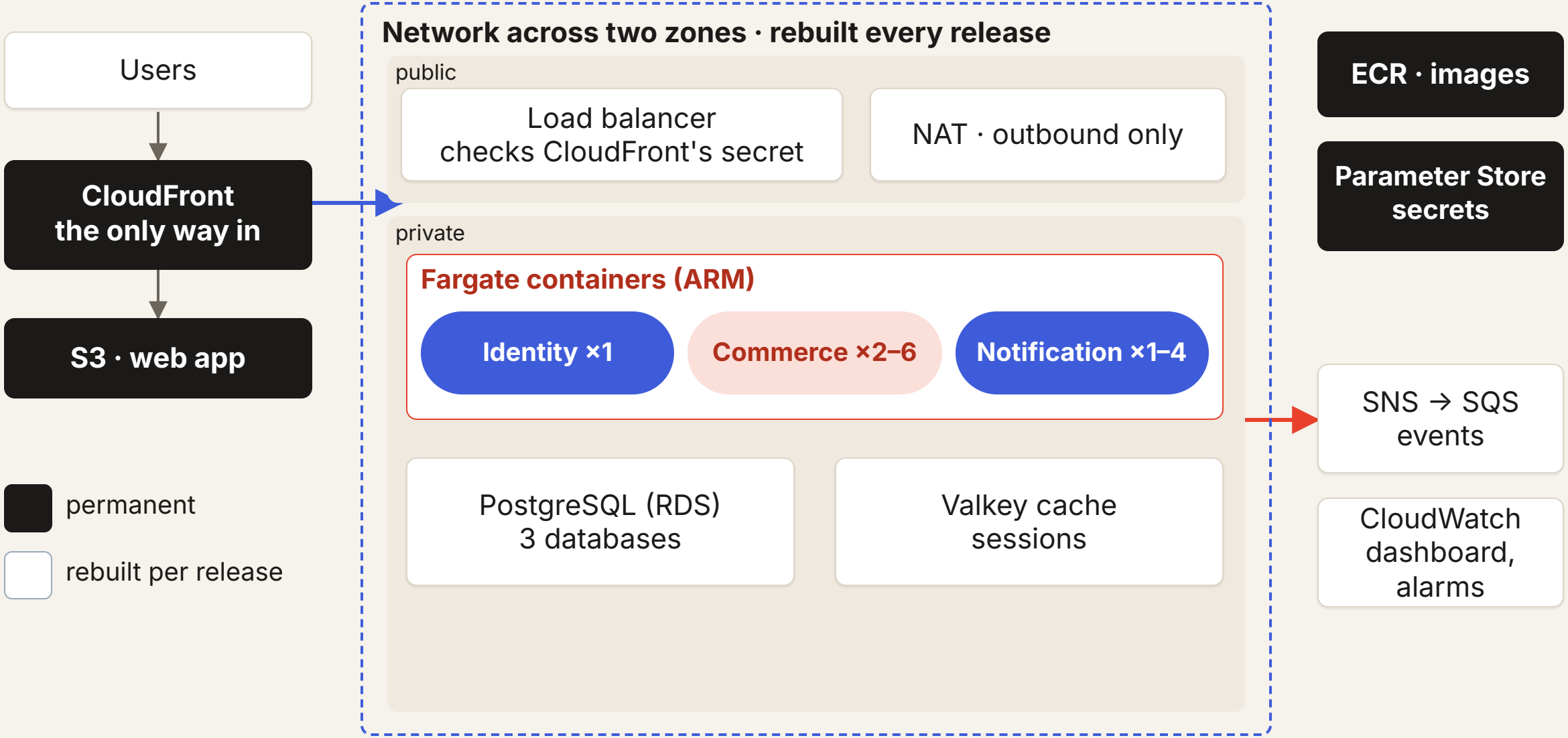
WZ Eric

Logical view, simplified



LJ Liu Jin

Physical view, simplified



YQ Yorik

Team expiry, numbered steps (fallback for slide 16)

